

Project Report on Data Visualization

“Crimes against Women – India”

Submitted by

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Contents

- 1. Abstract**
- 2. Dataset Description**
- 3. Python Libraries Used**
- 4. Data Visualization Charts**
- 5. Insights Gained**
- 6. Conclusion**

Abstract:

According to scroll.in, 26 crimes against women are reported every hour or one complaint every two minutes. Globally 81,000 women and girls were killed in 2020, around 47,000 of them (58%) died at the hands of an intimate partner or a family member. With this project our main objective is to throw light on the crimes against women in India with the help of graphs and to analyze which crime is highly reported. To show how crimes against women were increasing from 2001 to 2014 and also to provide insights on which state is safe for women to visit/travel.

Our Specific objective includes to find the State/UT with the highest and lowest crime rate.

Dataset Description:

The Dataset used for the project has state-wise and district level data on the various crimes committed against women between 2001 to 2014. Crimes that are included are Rape, Kidnapping and Abduction, Dowry Deaths, Assault on her Modesty, Insult to her Modesty, Domestic Violence by Husband/Relatives and Importation of Girls.

	STATE/UT	DISTRICT	Year	Rape	Kidnapping and Abduction	Dowry Deaths	Assault on women with intent to outrage her modesty	Insult to modesty of Women	Cruelty by Husband or his Relatives	Importation of Girls
0	ANDHRA PRADESH	ADILABAD	2001	50	30	16	149	34	175	0
1	ANDHRA PRADESH	ANANTAPUR	2001	23	30	7	118	24	154	0
2	ANDHRA PRADESH	CHITTOOR	2001	27	34	14	112	83	186	0
3	ANDHRA PRADESH	CUDDAPAH	2001	20	20	17	126	38	57	0
4	ANDHRA PRADESH	EAST GODAVARI	2001	23	26	12	109	58	247	0

Dataset URL - <https://www.kaggle.com/datasets/greeshmagirish/crime-against-women-20012014-india>

The Dataset was inconsistent. Therefore, Data Preprocessing was carried, many duplicate values were present in the State/UT column, few columns were not necessary for the further process, however there were no null values. After Data Preprocessing, we found the dataset more consistent and easier to work on.

The dataset consisted the data from 2001 -2014, on crimes committed in 29 States and 6 Union Territories. It consisted data on 954 unique districts. Below is the output on information on dataset before and after preprocessing.

Before Data Preprocessing

```
dataset.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10677 entries, 0 to 10676
Data columns (total 11 columns):
 #   Column                                     Non-Null Count  Dtype
---  -
 0   Unnamed: 0                               10677 non-null  int64
 1   STATE/UT                                 10677 non-null  object
 2   DISTRICT                                 10677 non-null  object
 3   Year                                     10677 non-null  int64
 4   Rape                                    10677 non-null  int64
 5   Kidnapping and Abduction                10677 non-null  int64
 6   Dowry Deaths                           10677 non-null  int64
 7   Assault on women with intent to outrage her modesty 10677 non-null  int64
 8   Insult to modesty of Women              10677 non-null  int64
 9   Cruelty by Husband or his Relatives      10677 non-null  int64
10   Importation of Girls                     10677 non-null  int64
dtypes: int64(9), object(2)
memory usage: 917.7+ KB
```

After Data Preprocessing

```
dataset.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 10186 entries, 0 to 10675
Data columns (total 10 columns):
 #   Column                                     Non-Null Count  Dtype
---  -
 0   STATE/UT                                 10186 non-null  object
 1   DISTRICT                                 10186 non-null  object
 2   Year                                     10186 non-null  int64
 3   Rape                                    10186 non-null  int64
 4   Kidnapping and Abduction                10186 non-null  int64
 5   Dowry Deaths                           10186 non-null  int64
 6   Assault on women with intent to outrage her modesty 10186 non-null  int64
 7   Insult to modesty of Women              10186 non-null  int64
 8   Cruelty by Husband or his Relatives      10186 non-null  int64
 9   Importation of Girls                     10186 non-null  int64
dtypes: int64(8), object(2)
memory usage: 875.4+ KB
```

Python Libraries Used:

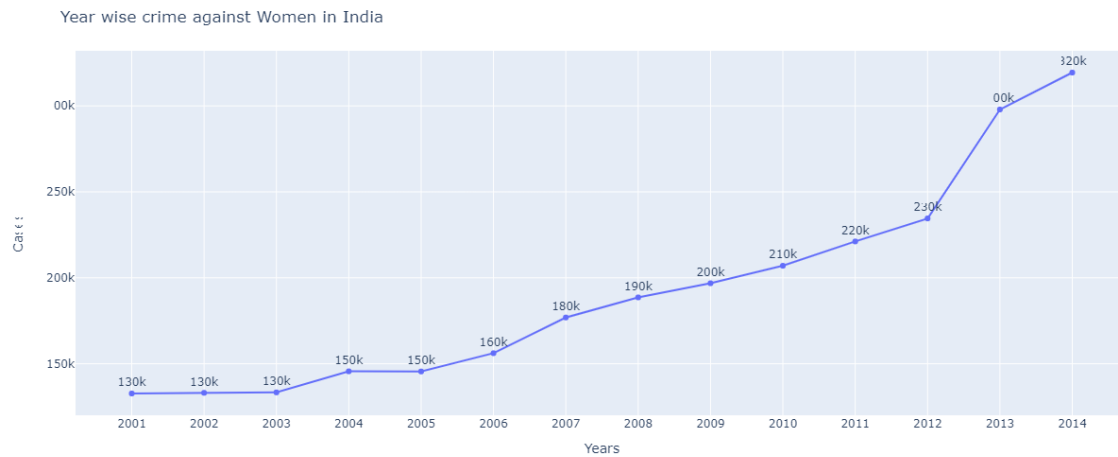
Importing necessary libraries

```
[2] import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import opendatasets as od
```

To make the visualization interactive and aesthetic we have used four libraries in the project. Namely, Pandas, Plotly.express, Plotly.graph_objects and opendatasets libraries.

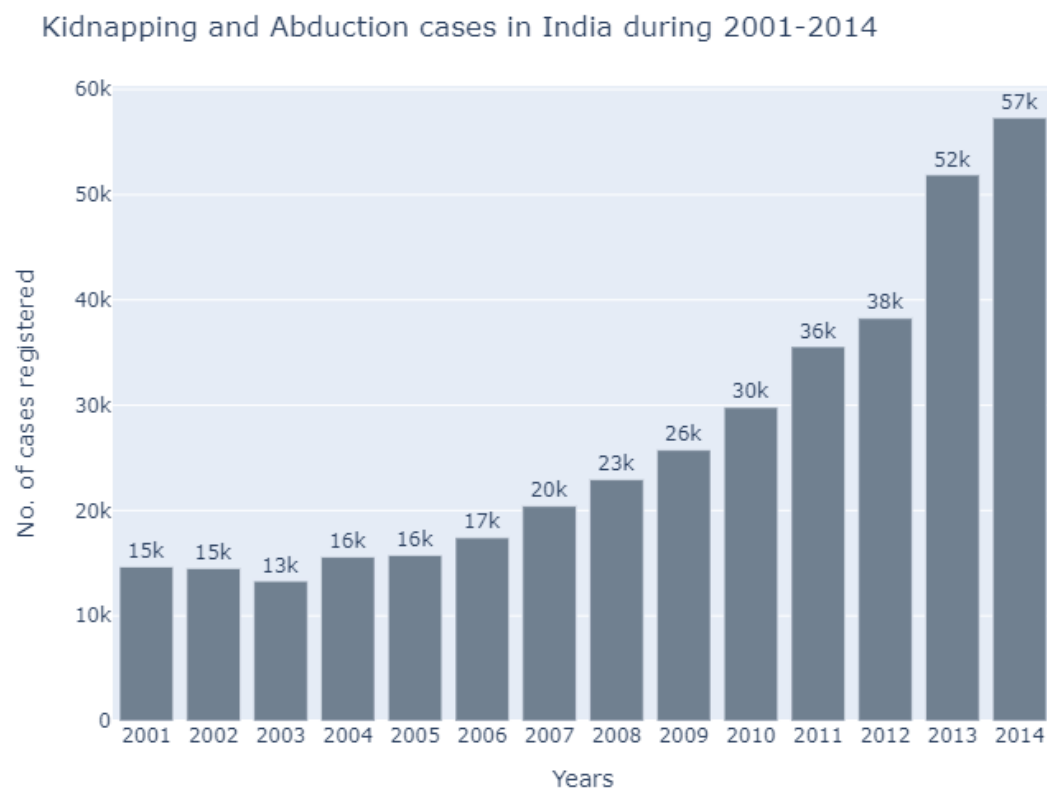
Pandas was used for data analysis and data manipulation. The Plotly library was used for the data visualization aspect and the opendatasets was used to access datasets from online resources.

Graphs and Charts Generated Using the Dataset:



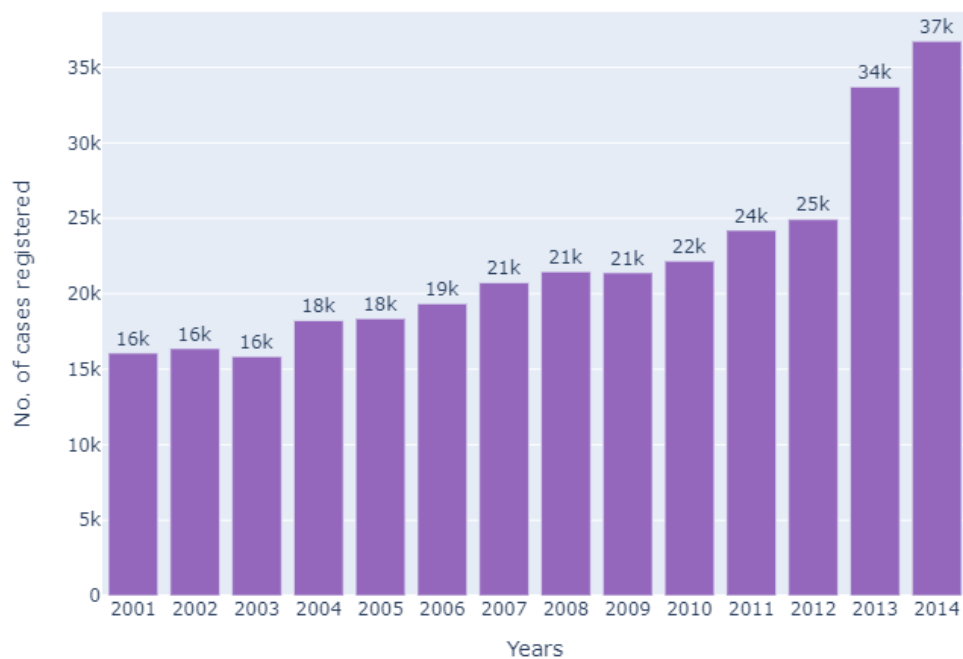
The above line plot shows the cases registered year wise from 2001-2014.

Now to get more insights on each crime we plotted bar graphs on year vs No. of cases registered for 14 years of data. Below are the results.



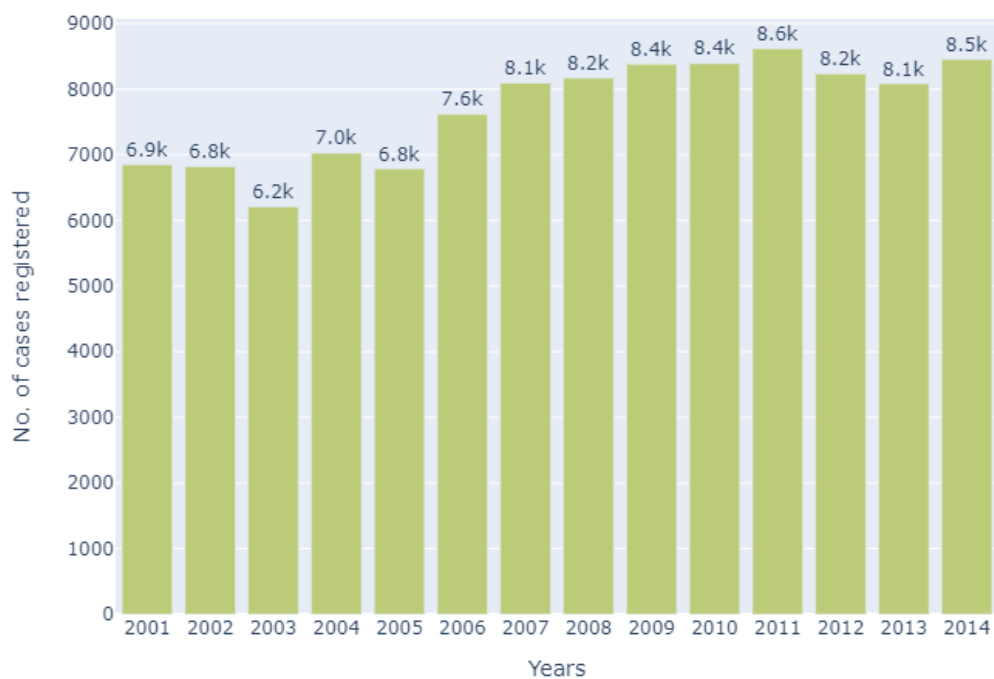
In 2014, we can see that the kidnapping and abduction cases are increased thrice than that of 2001.

Rape cases in India during 2001-2014



In 2014, we can see that the Kidnapping and Abduction cases were increased thrice than that of 2001.

Dowry Deaths in India during 2001-2014



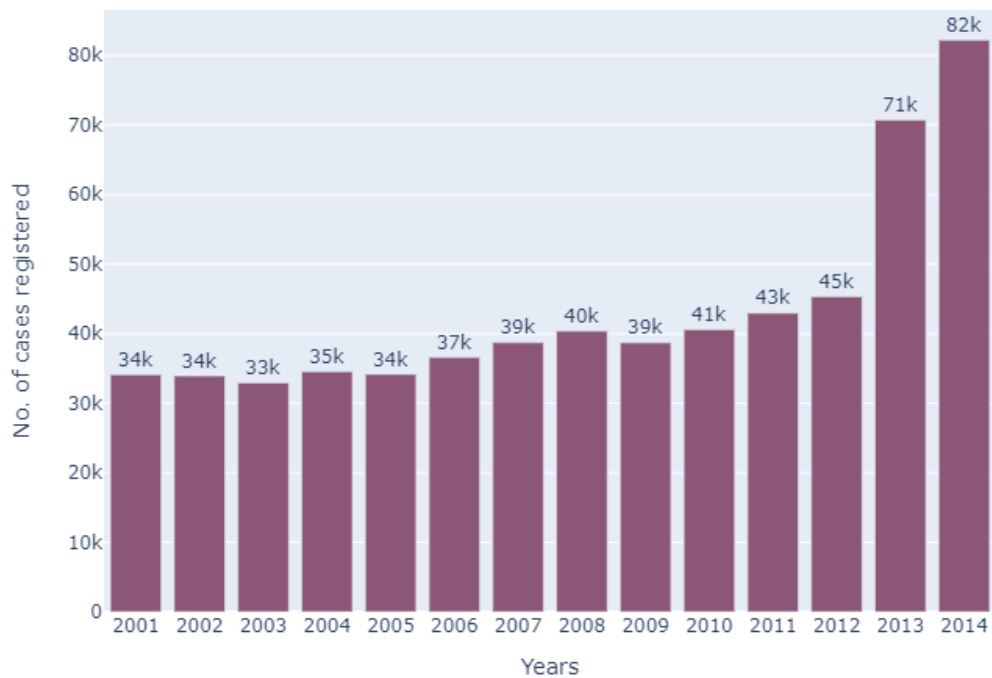
Though the literacy rate of India is increased, there is not much change in dowry cases.

Domestic Violence cases in India in 2001-2014



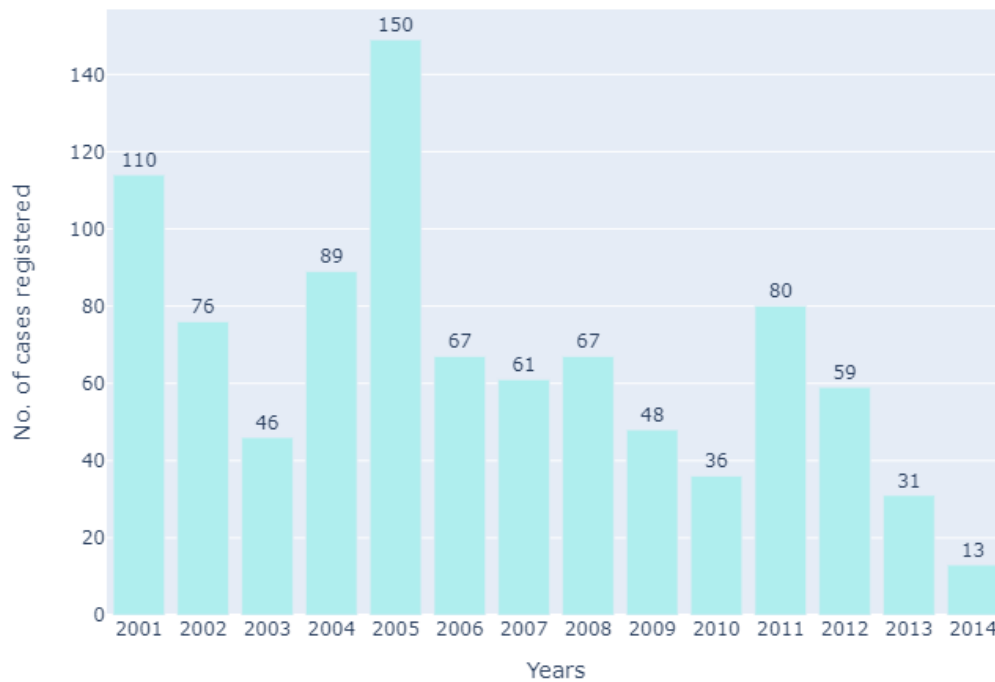
The Domestic Violence cases are 60% more in 2014 than in 2001.

Assault to her modesty in 2001-2014



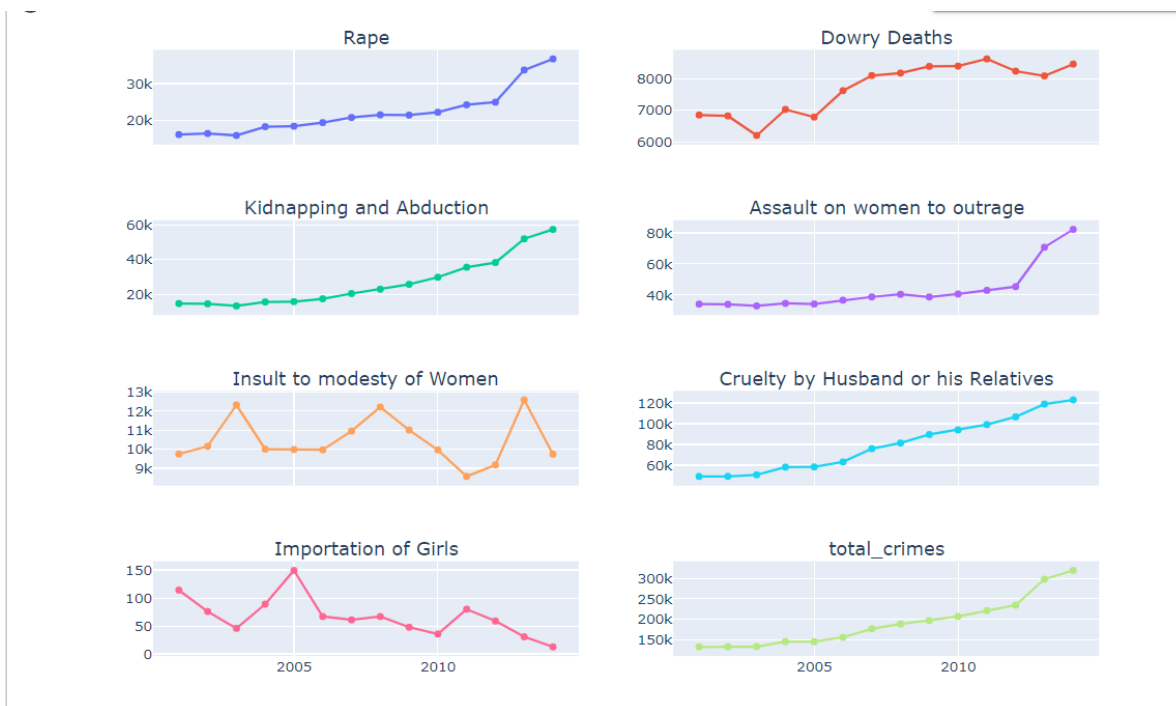
Assault on women is increased by 60%.

Trafficking of Girls during 2001-2014



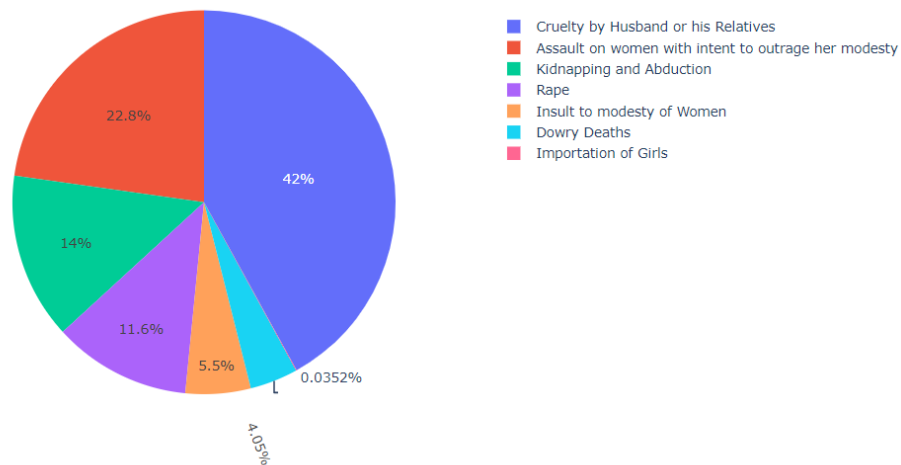
Female Trafficking is the only crime which had reduced considerably. There was a reduction of 88% from 2001 to 2014.

When the crimes were plotted next to next, we can more clearly observe the patterns of which crime is increasing consistently, decreased and fluctuating.



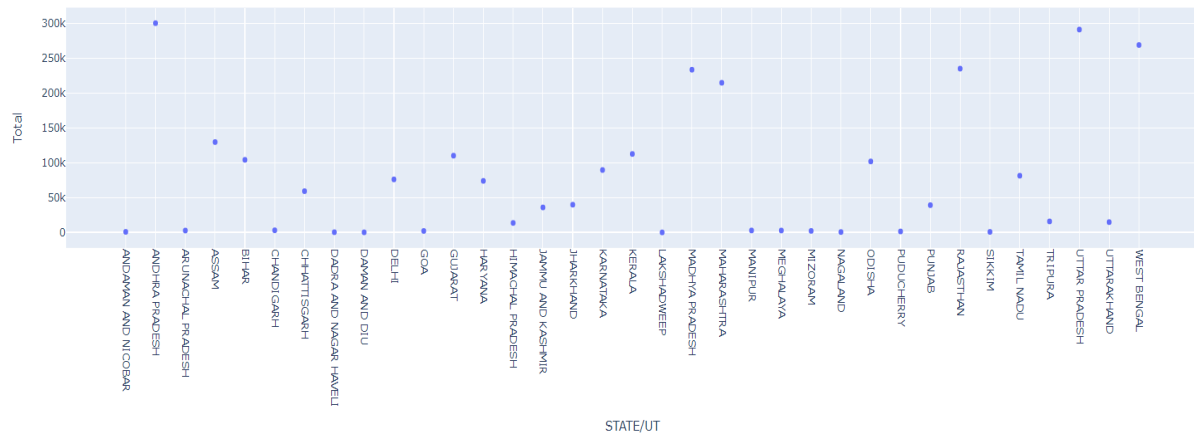
Further we can see the percentage on crimes reported using the pie chart.

Percentage of Each Crime between 2001 - 2014



Domestic Violence is the highest reported crime and female trafficking is the least reported.

Lastly, we tried to analyse data state wise with a scatter plot by plotting total crimes versus State/UT to find which State/UT has recorded highest and lowest crime rate.



According to this plot, the highest and lowest crime rates are found in Andhra Pradesh and Nagaland in states and Delhi and Lakshadweep in Union Territory respectively.

Insights gained:

1. Top 10 states with highest and lowest crime rate.

Total Cases		Total Cases	
STATE/UT		STATE/UT	
ANDHRA PRADESH	300355	LAKSHADWEEP	27
UTTAR PRADESH	291199	DAMAN AND DIU	115
WEST BENGAL	268988	DADRA AND NAGAR HAVELI	283
RAJASTHAN	235093	NAGALAND	536
MADHYA PRADESH	233543	SIKKIM	741
MAHARASHTRA	214792	ANDAMAN AND NICOBAR	784
ASSAM	129682	PUDUCHERRY	1359
KERALA	112588	GOA	2062
GUJARAT	110153	MIZORAM	2153
BIHAR	104188	ARUNACHAL PRADESH	2656

2. **Andhra Pradesh** had registered the most crimes against women from 2001 to 2014.
3. **Madhya Pradesh** reported the highest number of Rape cases and Assault on her modest cases.
4. **Uttar Pradesh** reported the highest number of Kidnapping and Abduction and Dowry Deaths.
5. **Madhya Pradesh, Andhra Pradesh, Uttar Pradesh, Rajasthan** and **West Bengal** account for almost half of all crimes committed against women around the country from 2001 to 2014.
6. We can find that except trafficking, all other crimes are increasing each year.
7. Finally, the highest and lowest crime rates are found in **Andhra Pradesh** and **Nagaland** in States and **Delhi** and **Lakshadweep** in Union Territory respectively.

Conclusion:

With this project we tried to analyze and visualize the dataset to know the crimes committed against women in India from 2001-2014. It was found that all crimes were increasing gradually except importation of girls. We have to realize this dataset was of only reported crimes, many cases go unreported due to family pressures or other reasons. And we can even infer that after 2014 there may even be seven times more reported cases than that of 2014. Because many awareness programs and laws were introduced for women safety which led many women file cases.

We should know that this was the data on one country, if we take it on a global scale, we can easily find that the most registered crime would be that which are done against women.